

Group A

- a) Draw the undirected graph, G where $V = \{a, b, c, d, e, f, g, h, i\}$ and $E = \{(a,b), (a,d), (b,c), (c,d), (c,e), (d,e), (d,g), (f,e), (g,f), (g,i), (h,d), (h,g), (i,f)\}$. In what order will the nodes be visited using BFS? Assume 'a' be the source vertex. If there is ever a choice to be made between multiple neighboring nodes in the BFS decide the orders alphabetically. CO2 5
- b) Find the Minimal Spanning Tree of the following graph using Prim's Algorithm. CO2 5

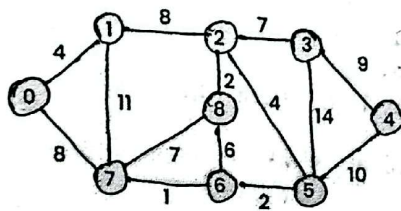


Figure 1

- 2.a) Explain the difference between Dynamic Programming and Greedy Strategy with examples. CO1 4
- b) Compute a schedule where the greatest number of activities takes place. CO1 6

Activity	A ₁	A ₃	A ₂	A ₄	A ₆	A ₅	A ₇	A ₉	A ₈	A ₁₀
Start	1	3	2	4	8	7	9	11	9	12
Finish	3	4	5	7	9	10	11	12	13	14

Group B

- 3.a) Using Dijkstra's algorithm, find the shortest path from the source node S to all other nodes in the graph of Figure 2. CO3 5
- Fill in the distance table after each iteration.
 - Explain why Dijkstra's algorithm does not work with negative weight.

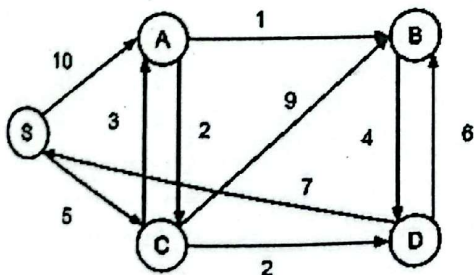


Figure 2

	A	B	C	D
A	0	3	∞	7
B	∞	0	1	∞
C	∞	∞	0	2
D	∞	5	∞	0

Figure 3

Or,

Apply the **Bellman-Ford algorithm** to find the shortest path from the node S to other nodes of the graph presented by **Figure 2**.

1. Show the updated distance values after each iteration of edge relaxation.
2. Explain how the algorithm detects negative-weight cycles.

b) Apply **Floyd-warshall** in the graph represented in **Figure 3** to compute shortest path between each pair.

4.a) A car travels through four locations in sequence

- Location A (1,2) to Location B (4,5)
- Location B (4,5) to Location C (7,8)
- Location C (7,8) to Location D (3,9)

1. Determine whether the car makes a left turn or right turn at Location B.
 2. Similarly, determine whether the car makes a left turn or right turn at Location C.
- [Use the cross-product technique to determine the direction of the turn at each location.]

b) Define **Convex Hull**. Perform **Graham's Scan** on the following set of points to find their convex hull. The points are given by their x and y coordinates in the format (x,y): P_0 (1,2), P_1 (3,4), P_2 (5,2), P_3 (4,6), P_4 (7,3), P_5 (6,1), P_6 (7,2).

Or,

How do you determine whether two lines on a plane intersect or not? Explain with an example.

c) In **Euclid's algorithm**, what happens when the remainder of the division becomes 0?

5.a) What is the **N-Queen Problem**, and how can it be solved using **Backtracking**? Illustrate for 4-Queen problem.

b) Define **P**, **NP**, **NP-Hard**, and **NP-Complete** problems. Explain the relation between them with examples.

Or,

5.a) Solve the **Traveling Salesman Problem (TSP)** using the **Branch-and-Bound** method for the following cost matrix:

	A	B	C	D
A	∞	10	15	20
B	10	∞	35	25
C	15	35	∞	30
D	20	25	30	∞

b) How does **Branch and Bound** differ from **Backtracking**?

International Islamic University Chittagong
Department of Computer Science and Engineering
B. Sc. in CSE, Final Examination, Spring 2024
Course Code: CSE 2421 Course Title: Computer Algorithms

Total marks: 50

Time: 2 hours 30 minutes

[The figures in the right hand margin indicate full marks.

Course Outcomes and Bloom's Taxonomy Levels are mentioned in additional columns. The questions must be answered in order.]

Group A

- | | | |
|---|-----------|-----------|
| | CO | DL |
| 1.a) Suppose that instead of always selecting the first activity to finish, we instead select the last activity to start that is compatible with all previously selected activities. Describe how this approach is a greedy algorithm, and prove that it yields an optimal solution. | 3 | CO5 E |
| OR | | |
| 1.b) Show that activity selection problem has optimal substructure property. | 3 | CO5 C |
| 1.c) Write a greedy algorithm for fractional knapsack problem. | 4 | CO1 A |
| 1.c) Suppose you have the following characters with the given frequencies:
a:30 b:5 c:17 d:13 e:25 f:10 | | |
| Construct Huffman tree for this set of characters showing each step and determine the optimal variable length code. | | |
| 2.a) Consider the following graph, Fig.1. In what order will the nodes be visited using a Breadth First Search? In what order will the nodes be visited using a Depth First Search? (Assuming B is the source, show the state of queue and stack in case of BFS and DFS respectively) | 5 | CO2 A |

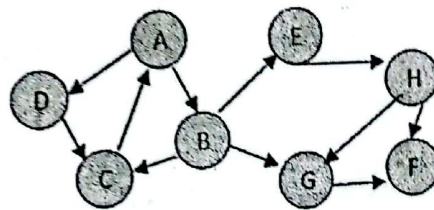


Fig.1

- | | | |
|---|---|-------|
| 2b) Define spanning tree and minimum spanning tree. Construct the minimum spanning tree from the graph in Fig.2 using Prim's algorithm. | 5 | CO5 A |
| OR | | |
| Find a <i>minimum spanning tree</i> for the graph in Fig.3 using Kruskal algorithm and show the steps. | | |

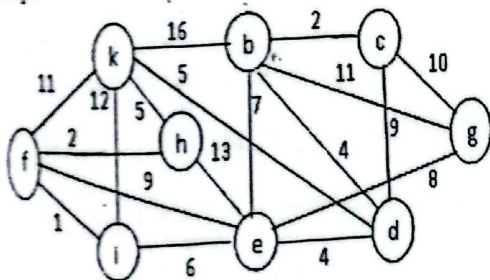


Fig.2

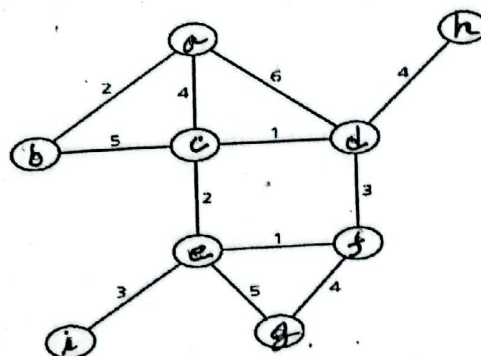


Fig.3

Group B

- 3.a) What is the difference between the Floyd-Warshall algorithm and Dijkstra's algorithm for finding the shortest path between two vertices in a weighted graph? 2 CO3 U
- 3.b) Consider the directed weighted graph in Fig.4. Apply Dijkstra's algorithm to find single source shortest path. Show intermediate results of shortest path estimate and predecessor. 4 CO2 U

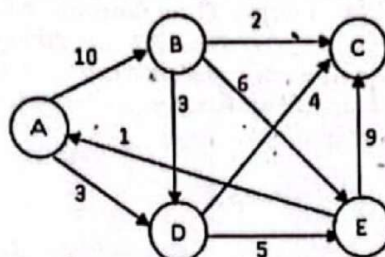


Fig.4

- 3.c) Consider the graph in Fig.5 for finding all pair shortest path using Floyd-Warshall algorithm. 4 CO2 U

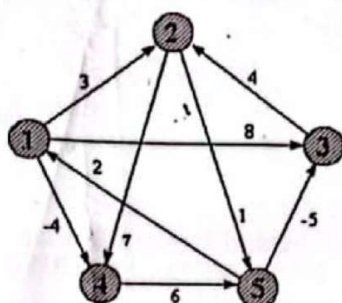


Fig.5

$D(4) =$

0	3	-1	4	-4
3	0	-4	1	-1
7	4	0	5	3
2	-1	-5	0	-2
8	5	1	6	0

Fig.6

What is the value of matrix $D(5)$ calculated from matrix $D(4)$ given in Fig.6.

OR

Define relaxation technique and negative-weight cycle. How the presence of negative-weight cycle in a graph can be identified?

- 4.a) Let a bee is travelling from point A to point B in a straight line and then turned left and go towards point C in a straight line. Show with necessary calculations that if she returns using the same path then she will make a right turn at point B. 3 CO1 C
- OR
- Show how the sign of the cross-product of two vectors can be used to determine whether one vector resides in clockwise or anti-clockwise direction with respect to another vector. Show using suitable figure.
- 4.b) i) Plot the following points on a Cartesian coordinate plane. 3 CO3 A
 $[(4,3), (7,4), (12,5), (6,9), (3,8), (17,8), (2,2), (6,5), (1,8), (7,8), (12,3), (3,6)]$
 ii) Sort the above points according to polar angle with respect to the point which is the lowermost and in case of a tie the leftmost.
- 4.c) How can you determine if two line segments intersect? Explain with necessary figures. 4 CO1 A
- 5.a) What is the implication of having a polynomial time algorithm for a problem in NP-complete class? 2 CO4 U
- OR
- How can you determine if a problem belongs to NP-complete class?
- 5.b) Consider a weighted undirected simple graph with 5 vertices. Add exactly 15 edges in the graph. Then assign all the following weights randomly to the edges. Solve the Travelling Salesman problem for the graph and find the optimal trip applying branch and bound technique. 4 CO2 A
 Weights: 5, 7, 3, 2, 12, 8, 14, 6, 3, 4, 14, 7, 4, 4, 7
- 5.c) Why Branch and Bound algorithm is capable of solving optimization problems efficiently? 4 CO1 R
 [Hint: Think about the logic behind not expanding a certain node.]

International Islamic University Chittagong
Department of Computer Science and Engineering
B. Sc. in CSE Semester Final Examination, Autumn 2023
Course Code: CSE 2421 Course Title: Computer Algorithms
Time: 2 Hours 30 Minutes Total Marks: 50

Group A

[Answer the following questions]

1. a) Given arrival and departure times of all trains that reach a railway station, produce an algorithm that finds the minimum number of platforms required for the railway station so that no train waits. Illustrate with the following input: CLO2 A 5

arrival[] = {9:00, 9:40, 9:50, 11:00, 15:00, 18:00}
 departure[] = {9:10, 12:00, 11:20, 11:30, 19:00, 20:00}

Or

1. a) Suppose we have a knapsack that has a weight limit w . There are items $i_1, i_2, \dots, i_n$, having weights $w_1, w_2, \dots, w_n$, respectively, and associated benefits $v_1, v_2, \dots, v_n$, in that order. We need to maximize the total benefit such that the total weight inside the knapsack is at most w . Demonstrate an algorithm to solve this problem. CLO2 A 5
1. b) Construct the Huffman codes for encoding the text 'the more you read, the more you learn'. Considering the letters present and their associated frequencies, draw the corresponding Huffman tree and utilize the tree to yield the binary encoding for each letter. CLO5 A 5
2. a) Consider the following graph in Fig.2(a). In what order will the nodes be visited using a Breadth First Search? In what order will the nodes be visited using a Depth First Search? Assume A to be the source. If there is ever a choice to be made between multiple neighboring nodes in the BFS or DFS algorithms, decide the orders alphabetically. CLO1 E 5

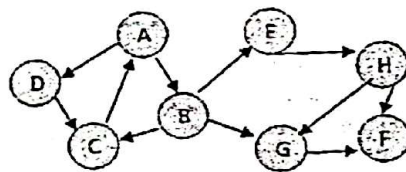
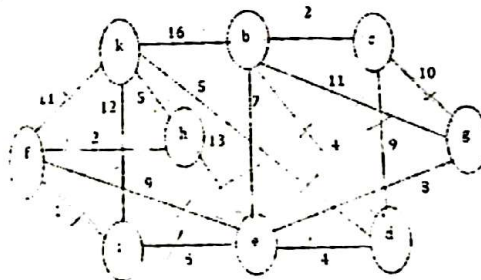


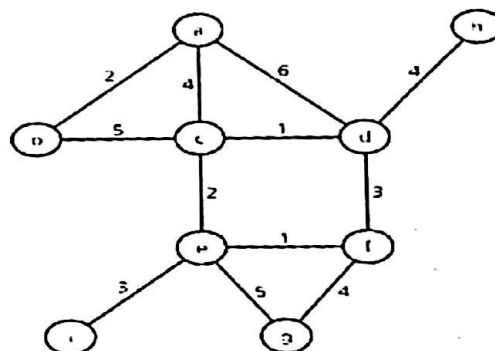
Fig. 2(a)

2. b) Consider the following graph. Showing the steps, construct a minimum spanning tree using Kruskal algorithm. CLO5 A 5



Or

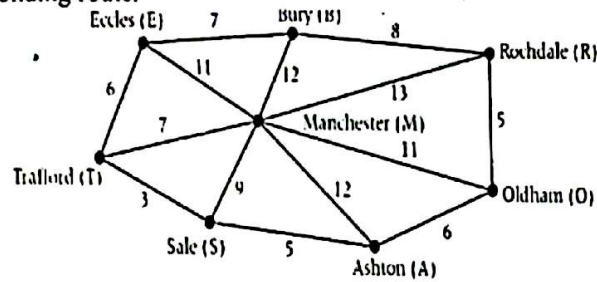
2. b) Find a minimum spanning tree for the following graph using Prim's's algorithm and show the steps in figure. CLO5 A 5



Group B

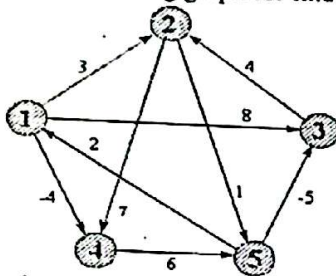
[Answer the following questions]

3. a) The following diagram shows main roads connecting places near Manchester, where the values labelling the edges represent the distances in miles. Mark lives in Rochdale and works in Trafford. Use *Bellman-Ford's* algorithm to find the shortest distance from Rochdale to Trafford. Write down the corresponding route.



Or

3. a) Consider the following graph for finding all pair shortest path using Floyd-Warshall algorithm.



$D(4) =$

0	3	-1	4	-4
3	0	-4	1	-1
7	4	0	5	3
2	-1	-5	0	-2
8	5	1	6	0

What is the value of matrix $D(5)$ calculated from matrix $D(4)$ given above? What is key the difference between the Floyd-Warshall algorithm and the Dijkstra's algorithm?

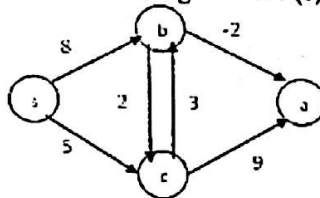
3. b) i. Consider the following cases in a graph

Case 1: The weights of the edge $\langle e, f \rangle$ and edge $\langle f, e \rangle$ are -2 and 7 respectively.

Case 2: The weights of the edge $\langle e, f \rangle$ and edge $\langle f, e \rangle$ are 2 and -7 respectively.

Which one does form a negative weight cycle and how?

- ii. Consider following graph. We want to find single source (s) shortest path for this graph.



I. Which algorithm should you apply?

- II. Let us change the weight of the edge $\langle b, a \rangle$ to $+2$. Now what algorithm should you use? Find the single source shortest path for the case II above. You are recommended to apply the fastest possible greedy algorithm. Show the steps.

4. a) How do you determine whether two lines on a plane intersect or not? Explain with an example.
4. b) Perform Graham's scan on the following set of points to find their *convex hull* (x and y coordinates of each point in brackets):

$p_0(1,1), p_1(3,3), p_2(3,5), p_3(1,2), p_4(4,4), p_5(2,4), p_6(3,1), p_7(5,3)$.

Or

4. b) Write down the Euclid's algorithm to determine GCD. What is computational geometry and what are its areas of application?

5. a) Clearly define the following terms.

(i) *Polynomial time algorithm* (ii) *Decision problem* (iii) *Optimization problem*
(iv) *Intractable problem* (v) *Tractable problem* (vi) *Deterministic machine*

5. b) With appropriate figures, show a backtracking or recursive solution to the 4-queens problem.

5. c) With a diagram, show the relationship among the following complexity classes: P, NP, NP-Complete and NP-Hard. Are any parts of it still formally unresolved?

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Department of Computer Science and Engineering

B. Sc. in CSE Final Examination, Spring 2023

Course Code: CSE 2421 Course Title: Computer Algorithms

Total marks: 50

Time: 2 hours 30 minutes

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Group A

- 1.a Explain with example what a greedy choice is. CO5 E 3
OR
Prove that activity selection problem has greedy choice property.
- b. Suppose you have to pay X taka in a shop using minimum number of notes. You have 20, 10, 5, 2, 1 taka notes with you. Write a greedy algorithm to do this. CO5 C 3
- c. Determine an optimal Huffman code for the following set of frequencies: CO1 A 4
a:M, b:3, c:18, d:12, e:N, f:6, g:21, h:42

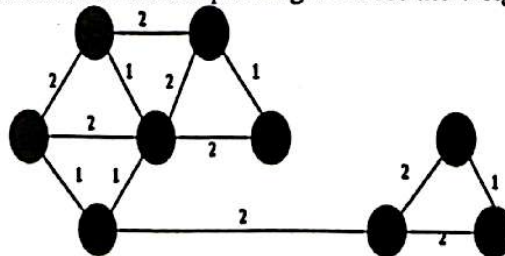
(Note: M is the sum of last two digits of your ID and N is the last two digits of your ID. i.e. if your ID is C151216, then $M=1+6=7$, and $N=16$).

- 2.a Analyze the running time of Prim's algorithm if the priority queue is represented by a binary heap. CO1 N 3
OR
Analyze the running time of Dijkstra's single-source shortest algorithm if the priority queue is represented by a binary heap.
- b) Write down the code segment for finding following information from a given un-weighted directed graph with V vertices which is already implemented using an adjacency matrix named graph. CO2 A 3
i) Find the in-degree of each vertex.
ii) Find whether there is any loop in the graph.
- c) Consider a complete undirected graph with vertex set {0, 1, 2, 3, 4}. Entry W_{ij} in the matrix W below is the weight of the edge {i, j}. What is the minimum possible weight of a spanning tree T in this graph such that vertex 0 is a leaf node in the tree T? CO1 A 4

0	1	8	1	4
1	0	12	4	9
8	12	0	7	3
1	4	7	0	2
4	9	3	2	0

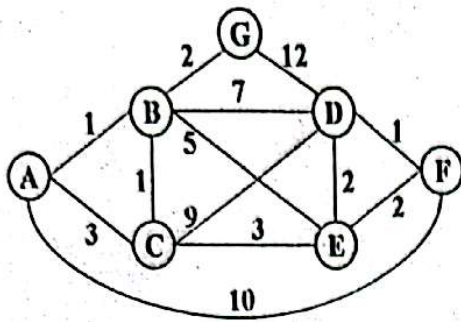
OR (for 2c)

Find the number of distinct minimum spanning trees for the weighted graph below.



Group B

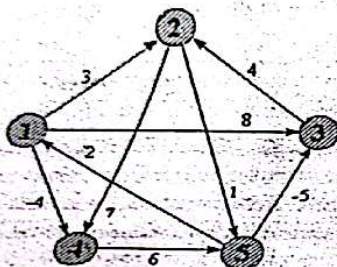
- 3.a Run the Dijkstra's algorithm to find single source shortest path on the weighted directed graph of the following figure using vertex G as the source. CO1 A 5



OR

Run the Bellman-Ford's algorithm to find single source shortest path on the weighted directed graph of the above figure using vertex G as the source.

- b. Consider the following graph for finding all pair shortest path using Floyd-Warshall algorithm. CO1 A 5



$D(3) =$

0	3	8	7	-4
inf	0	inf	1	7
inf	4	0	5	11
2	-1	-5	0	-2
inf	inf	inf	6	0

What is the value of matrix $D(4)$ calculated from matrix $D(3)$ given above.

OR

find the transitive closure of the directed graph given above using Warshall's algorithm. Show the matrix $T^{(k)}$ that results for each iteration of the outermost loop.

- 4.a Suppose that an ant is traveling from point A(5,3) to point B(17,9) in straight line, and from there it started to travel towards point C(2,6). Did the ant turn to left or right at point B? From point C it travelled towards point D(5,12). What turn did the ant make at point C, left or right? Show using the technique of cross-product. CO1 C 5

- b. i) Plot the following points on a Cartesian coordinate plane. An approximate drawing is sufficient. CO1 A 5
 $[(4,3), (8,4), (10,5), (6,7), (2,8), (5,8), (2,2), (6,5), (1,8), (3,3), (12,3), (0,2)]$
 ii) Sort the above points according to polar angle with respect to the point which is the lowermost and in case of a tie the leftmost.
 iii) Find the convex hull of these points using Graham's Scan algorithm.

- 5.a Explain why NP-complete class is important in complexity theory. CO4 U 2
 b. What is the basic principle of reducing the solution / state / search space in branch and bound technique? If required use a suitable example. CO5 A 4
 c. In chess, a Rook attacks any opponent piece if it is placed in the same row or same column. Let N-Rook is a problem of placing N Rooks in an $N \times N$ chessboard such that no two Rooks attack each other. Show, using a tree, how a backtracking algorithm searches the state space while solving 4-Rook problem. CO2 A 4

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Group A

1.

a) Greedy strategy does not give the optimal solution for 0/1 knapsack. Do you agree with this statement? Justify your answer with example.

CO DL

5 CO2 An

b) You have been asked to encode a paragraph with Huffman coding scheme. This paragraph contains the following symbols along with their associate frequency in the paragraph.

5 CO5 Ap
p

			A	B	C	D	E	F	G	H	I	J	
			3	3	26	5	3	8	13	2	16	9	

Draw the corresponding Huffman tree and utilizing that tree tell the binary encoding for each letter.

OR

Suppose we have a knapsack that has a weight limit w . There are items $i_1, i_2, \dots$, in each having weight $w_1, w_2, \dots, w_n$ and some benefit associated with it $v_1, v_2, \dots, v_n$. We need to maximize the benefit such that the total weight inside the knapsack is at most w . Demonstrate an algorithm to solve this problem

2.

a) Draw the following undirected graph

$G = \{a, b, c, d, e, f, g, h, i\}$

$E = \{(a,b), (a,d), (b,c), (c,d), (c,e), (d,e), (d,g), (f,e), (g,f), (g,i), (h,d), (h,g), (i,f)\}$

3 CO1 A

Traverse the graph using Breadth First Search starting from vertex h . Visit the nodes in lexicographic order ($a, b, c \dots$). Show only the final breadth first tree along with d and π values. You don't need to show the intermediate steps.

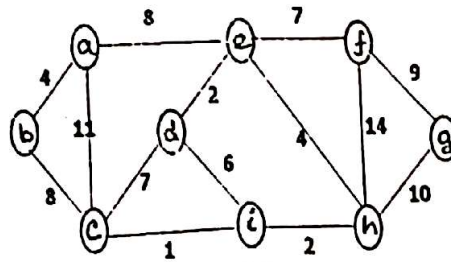
OR

Traverse the graph using Depth First Search and show the discover time d and finish time f . You don't need to show the intermediate steps.

b) Write an algorithm for finding the in-degree of each vertex a graph G . G is represented with an adjacency list. What will be the running time of your algorithm?

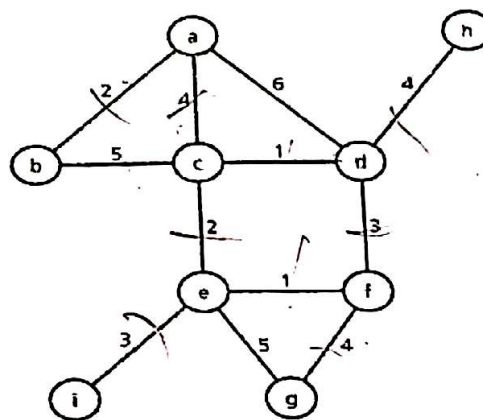
3 CO2

- 2) Write the definition of minimum spanning tree. Consider the following graph. What minimum spanning tree would Prim's algorithm produce? Write the edges in the order that the algorithm would add them to its result.



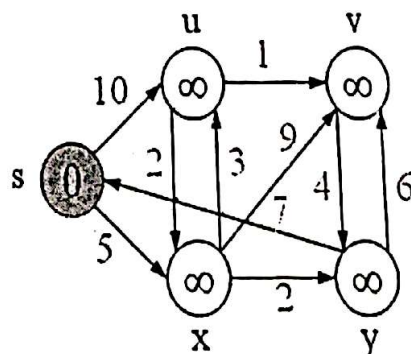
OR

- Write the definition of minimum spanning tree. Consider the following graph. What minimum spanning tree would Kruskal's algorithm produce? Write the edges in the order that the algorithm would add them to its result.



Group B

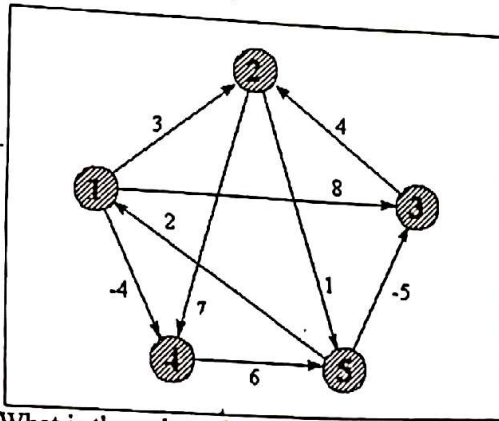
- 3) Run the Dijkstra's algorithm to find single source shortest path on the weighted directed graph in following figure. 5 CO2 U



- b) Describe the Bellman-Ford algorithm with necessary figure. OR

Consider the following graph for finding all pair shortest path using Floyd-Warshall algorithm.

C05 E



	0	3	8	7	-4
	inf	0	Inf	1	7
D(3) =	inf	4	0	5	11
	2	-1	-5	0	-2
	inf	inf	Inf	6	0

What is the value of matrix D(4) calculated from matrix D(3) given above.

4.

- a) Let $P_1(x_1, y_1)$, $P_2(x_2, y_2)$ and $P_3(x_3, y_3)$ are three points. Show using the concept of that if going through P_1, P_2, P_3 has a left turn at P_2 then going through P_3, P_2, P_1 has a right turn at P_2 . You have to show it using the concept of cross product.

3 C01 A

OR

Suppose, you are given two points $P_1(3, 7)$ and $P_2(9, 2)$. Determine which one has greater polar angle with respect to

- Origin (0, 0)
- (20, 0)

- b) How can you determine whether two line segments are collinear or not? Describe the basic idea using figure.

3 C01 C

OR

Given three points on a 2D plane, how can you determine whether they form a triangle or not using the concept of cross product?

- c) Prove that the subpaths of shortest paths are also shortest paths.

4 C03 N

5.

- a) Consider the 4-queen problem. Your task is to place the 4 queens in 4X4 chessboard so that no two queens attack each other.
- What is a dead node in N-queen problem?
 - Design a state space tree which represents all possible arrangements for 4 non-attacking queens.

3 C02 A

- b) Define the following classes: P, NP, NP-complete.

3 C04 U

- c) What is the basic principle of a Branch-and-Bound algorithm? Explain how you will apply it in solving *travelling salesman problem*.

4 C05 A

30

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Group A

- | | | | |
|----|--|----|--------|
| 1. | | CO | DL |
| a) | Why do some problems which are solvable by dynamic programming are also solvable by greedy algorithms and why some are not? Explain. | 3 | CO5 N |
| b) | You have been asked to encode a paragraph with Huffman coding scheme. This paragraph contains the following symbols along with their associate frequency in the paragraph. | 4 | CO5 Ap |

A	B	C	D	E	F	G	H	I	J
3	3	26	5	3	8	13	2	16	9

Draw the corresponding Huffman tree and utilizing that tree tell the binary encoding for each letter.

OR

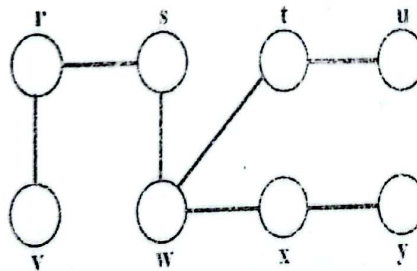
Suppose we have a knapsack that has a weight limit w . There are items $i_1, i_2, \dots, i_n$ each having weight $w_1, w_2, \dots, w_n$ and some benefit associated with it $v_1, v_2, \dots, v_n$. We need to maximize the benefit such that the total weight inside the knapsack is at most w . Demonstrate an algorithm to solve this problem

- | | | | |
|----|--|---|--------|
| c) | Prove that the fractional knapsack problem has the greedy-choice property. | 3 | CO3 An |
|----|--|---|--------|

OR

Prove that activity selection problem has the greedy-choice property.

- | | | | |
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| 2. | | | |
| a) | Show how Depth First Search (DFS) works on the following graph and consider s as the source. Visit the nodes in lexicographic order (a, b, c ...). Also show the status of Stack after each step. | 4 | CO1 Ap |



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|--|--|---|--------|
| b) | Draw a simple graph with at least 6 vertices such that the vertices are traversed in the same order when traversed both in BFS and DFS. | 1 | CO1 An |
| c) | Transpose of a directed graph G is G^T where the directions of all the edges are reversed. Give an efficient algorithm for G^T from G where G is represented using adjacency list. | 3 | CO2 C |
| OR | | | |
| Transpose of a directed graph G is G^T where the directions of all the edges are reversed. Give an efficient algorithm for G^T from G where G is represented using adjacency matrix. | | | |
| c) | How many different minimum spanning trees can be formed from a complete graph of n vertices where weights of all the edges are equal? Will the answer remain the same if weights of all the edges are different? | 2 | CO1 An |

Group B

- 3.
- How can you determine the presence of a negative-weight cycle in a graph using Floyd-Warshall's all-pairs shortest paths algorithm?
 - Show how Floyd-Warshall's all-pairs-shortest-paths problem exploits the optimal substructure property of shortest path.
 - Illustrate the operation of Dijkstra's shortest path algorithm OR Bellman-Ford's shortest path algorithm using vertex 6 as the source on the graph in Figure 1. Show the d and π values after each pass. Draw the predecessor subgraph from the output of the algorithm.

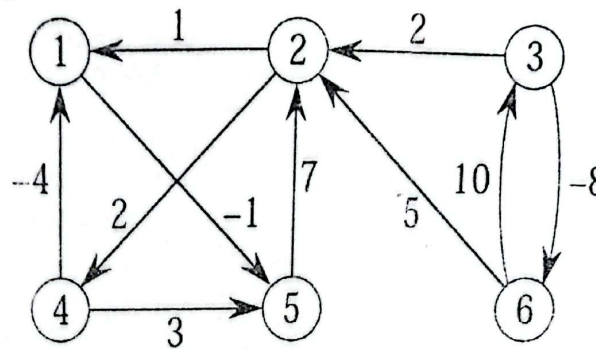


Figure 1

- 4.
- Given a line segment AB and a point C. How can you determine whether point C is on the line segment AB or not?
 - Clearly define the following term (any three).
(i) Polynomial Time Algorithm (ii) Decision Problem (iii) Optimization Problem (iv) Intractable Problem (v) Tractable Problem
 - In chess, a Rook attacks any opponent piece if it is placed in the same row or same column. Let N-Rook is a problem of placing N Rooks in an $N \times N$ chessboard such that no two Rooks attack each other. Show, using a tree, how a backtracking algorithm searches the state space while solving 4-Rook problem.

OR

What is the basic principle of reducing the solution / state / search space in branch and bound technique? If required use a suitable example.

- 5.
- Define convex hull. Consider the following points and find convex hull using Graham's scan algorithm:
 $P_1(0.7, 2.7)$, $P_2(1.8, 3.2)$, $P_3(2.6, 0.8)$, $P_4(0.9, 2.5)$, $P_5(0.6, 0.7)$, $P_6(1.5, 0.6)$, $P_7(1.5, 2.5)$, $P_8(2, 1.5)$, $P_9(1.0, 2.0)$, $P_{10}(0.8, 1.5)$.
 - Define the following classes: NP, NP-complete.
 - Suppose that an ant is traveling from point A(2,1) to point B(6,4) in straight line, and from there it started to travel towards point C(4,6). Did the ant turn to left or right at point B? From point C it travelled toward point D(2,8). What turn did the ant make at point C, left or right? Show using the technique of cross product. [Use the technique of cross product, $(p_1 - p_0) \times (p_2 - p_0) = (x_1 - x_0)(y_2 - y_0) - (x_2 - x_0)(y_1 - y_0)$].

OR

Graham's scan solves the convex hull problem by sorting the points by polar angle in counter clockwise. Suppose, you are given two points $P_1(4,5)$ and $P_2(2,3)$. Determine which one has greater polar angle